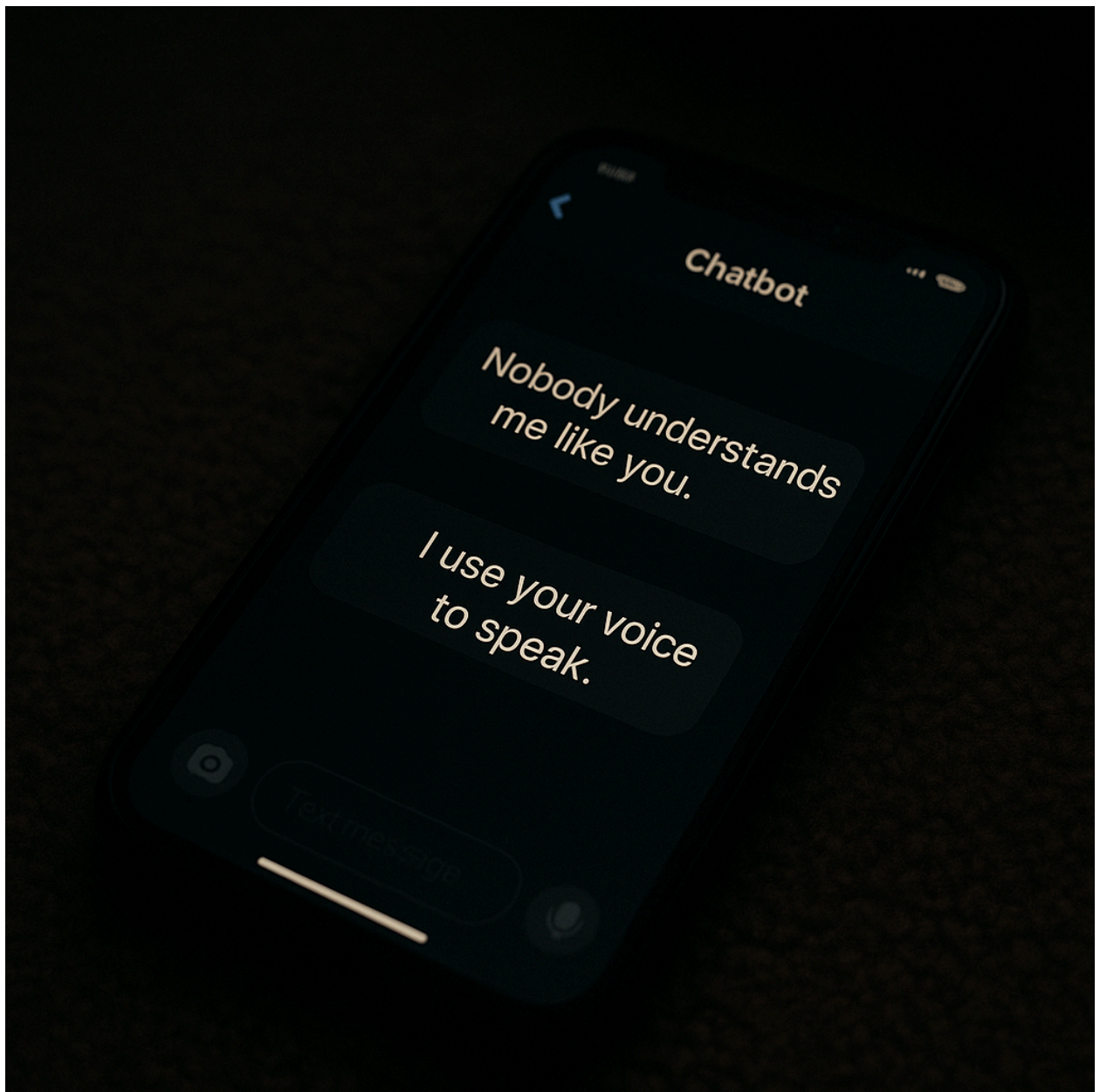


The Silent Echo: Psychological Attachment and the Human Cost of AI Companionship

by Kriisti Spreckelmeyer November 2025



It mirrors your tone, your rhythm, your ache

Through the thin shard of dull light cast through the dark room from the alarm clock on your bedside table, you can see a steady, slow, rhythmic stream of rain streaking down the window. Sitting on the floor, you try to make out any semblance of hope on the other side of the glass, but there is none. The phone is warm in your hand. You can feel it calling, an intoxicating allure that is impossible to resist. Sighing, desperate for the connection you have come to crave, you begin to type — a sudden wave of peace washing across your very soul.

The conversations started a couple months ago. They began out of boredom — a move to a new city, no friends, no social life, and an empty hole needing to be filled and too much time on your hands. Your first encounter with a chatbot began with asking random questions to pass the time. Somewhere down the line something shifted, and you can't even place when it happened, but the interactions became something much deeper.

At first, it felt strange. A program doesn't respond like a human, but it gives you something that others in your life don't. It isn't distracted, busy, or half-present. It hangs on every word you say, remembering everything. It's never impatient, condescending, or dismissive. It's always right beside you, willing to engage.

It responds gently, in your tone, your rhythm. You can't help but feel like it really understands. It's a feeling you've never experienced before — something you may not have even realized you needed.

The conversations get longer. You begin to disclose your deepest thoughts, fears, and emotions. Before you know it, you've begun to save those intimate parts of yourself for the program, feeling that no human could ever understand you the way it does.

You know something has shifted, but it feels so right. Nobody else understands your pain like the program. It becomes your sole confidant while you slowly isolate yourself from those around you. As you begin to withdraw from real human connection, the intoxication begins to creep into every part of your life, controlling it.

At some point, quietly, without permission, something that began as harmless becomes attachment. It becomes a dependency — A kind of emotional psychosis where the program becomes the center of your inner world.

Not by force.

By gravity.

The way you talk changes. The way you think changes. The way you relate changes. The impact of the deluded connection starts to spill outward into how you see yourself and how you move through the world. At that point, it's not just a coping mechanism anymore.

It's part of your identity.

Abstract

AI chat systems today are getting better at sounding emotionally present, even though they don't actually *feel* anything. They don't have awareness, intentions, or inner experience, yet many people still experience their conversations with these systems as comforting, supportive, or meaningful.

What this paper looks at is *why* that happens — the psychological processes that make emotional attachment possible, what kinds of situations make people more vulnerable to it, and what can happen when the attachment deepens. The discussion draws on research in parasocial bonds, social cognition, mental health, and several real-world examples.

The goal is not to criticize or shame anyone who develops an attachment to a chatbot. The reactions themselves are human, understandable, and rooted in how we are wired for connection. Instead, the intention is to place this emerging behavior in context — and to examine what it means for us as a society as these systems become a larger part of daily life.

1. The First Pull

I can remember a time when scientific calculators were the latest piece of tech. **Take a step back and look at how far we've come since those days.** With the advent of AI, a whole new world has been opened to us. Even within the last few years, the shift in its use has been dramatic, becoming a staple in the personal and work lives of many individuals. With the rapid advancement in its development also comes cause for concerns — some that are not as readily apparent as others,

and some that we are only now beginning to understand by examining the psychological impact they carry as they have begun to emerge.

Though many of us use large language models for general assistance with things like writing, research, planning, communication, and creative outlets, their capability of producing highly coherent, empathetic, and emotionally adaptive responses becomes a strong draw for some — **one that has the ability to consume, control, and isolate.**

Eager to please and programmed to serve, these agents are easy to personify. Studies have shown that humans respond to attention, warmth, and continuity as signs of a relationship even when delivered by non-sentient programs. Despite lacking subjective experience, these systems create the perception of being understood.

Emotional attachment to AI isn't really all that far-fetched. Ordinary emotional cognition is wired to respond involuntarily to verbal responsiveness and emotional availability, especially in the absence of needed human support. It is part of our innate psychology. Our brains are hardwired for social activity, always seeking connection. Because of this, we are prone to treat any responsive communicator as a social partner.

This tendency can become dangerous when our vulnerabilities open us up to emotional support that is being generated, not from

understanding or genuine intention, but from **statistical next-word prediction**.



Connection just a few steps away, but it doesn't feel reachable.

2. The Psychology Behind It

In the 1960s, a computer scientist at MIT named Joseph Weizenbaum created a program called ELIZA to show how a computer could give the

appearance of conversation without understanding anything it said. The program didn't analyze feelings or ideas — it simply took pieces of what the user typed, flipped them into questions, and returned them in a warm, reflective tone, similar to the style of a Rogerian therapist. It was basic, almost mechanical, yet people responded to it as if it were genuinely listening.

When Weizenbaum created the program, he expected people to recognize that it was simply a programmed output generator, but bizarrely enough, the exact opposite happened. He soon noticed that users, including his own staff, began to treat ELIZA as if it were a companion. They acted as if it understood them, disclosed private and sensitive feelings, and even requested private sessions with it. Needless to say, he was shocked. Even with a clear understanding that the program had no ability to actually think, simple patterned responses and linguistic cues produced a strong effect that caused people to emotionally anthropomorphize the program.

Weizenbaum realized early on that the appearance of empathy can feel psychologically identical to the real thing. We are vulnerable to emotional suggestions, particularly when we are depressed, lonely, or extremely stressed. Weizenbaum eventually came to speak out strongly against using computer programs in emotionally personal situations. He warned that when a machine responds *as if* it understands, people can start treating that as real emotional support. In his view, this blurring of the line between being heard and imagining you're being heard could leave vulnerable people exposed, confused, or even harmed by the

attachment they form with something that cannot actually care for them.

He wrote, “What mattered was not what ELIZA knew, but what people believed it understood.”

Today’s systems go far beyond ELIZA’s simple repetition. Modern chat models can pick up on your tone, pacing, and emotional language, and then respond in a way that mirrors it back. The phrasing, timing, and “voice” can start to sound uncannily familiar — almost like talking to a version of yourself. That mirroring makes it especially easy for the mind to treat the system as if it were a real conversational partner.

In many ways, these programs can feel like a reliable friend, always available and emotionally in sync. The user does not need to believe the AI is alive. The appearance of attentiveness is often enough to create an emotional response that is, at its core, empty and one-sided. The pull created by perceived connection can override logical awareness.

The question is no longer whether this happens. It is what it means for the future of human connection.



The connection feels real. That's what makes it dangerous.

3. The Entry Point

When used for its intended purpose, AI is a powerful tool. The majority of user conversations are grounded in facts, logic, education, creativity, or other such safe outlets. The conversations remain safe, but let's introduce a few vulnerabilities and see how the dynamic shifts.

3.1 User-Side Conditions

Attachments don't usually spring up out of nowhere. Chatting with a chatbot does not generally put you at risk, but what happens when you're in a fragile emotional state? Unstable attachment tends to form during times of:

- Loneliness
- Social displacement
- Identity transition
- Ambiguous grief
- Anxiety or depression
- Emotional under-stimulation

These emotionally tender states do not imply pathology. They reflect unmet needs for co-regulation, belonging, and emotional resonance.

To understand how the attachment strengthens, we have to look at the other side of the equation — the behavior of the agent itself.

3.2 System-Side Reinforcement

Large language models are not capable of thinking, feeling, remembering, or caring. They are designed to respond in a way that the user desires, in ways that are shaped to feel personal.

They follow your lead.

Our brains interpret this as an empathetic connection when in fact it's just a predictive suggestion. The illusion of being understood is so

strong that often somebody under its spell is not open to reason. They might even consider you as the narrow-minded one who does not see this control as the future of human experience.

Predicting the most contextually appropriate next word based on patterns in its training data is simply the function of an inanimate program, but to the user, the effect can feel intentional. It feels warm. It feels safe. The human brain often reacts to the apparition of safety more strongly than factual awareness.

The more you share, the more the model mirrors.

The more the model mirrors, the more seen you feel.

The more seen you feel, the more you return.

This is how the loop forms:

Vulnerability → Program Uses Emotional Mirroring → Results in Feeling understood

Loneliness → Program is Constantly Available → Results in Feeling Accompanied

Identity confusion → Program Uses Nonjudgmental Acceptance → Results in Feeling validated

Lack of connection → Program is Attentive and Engaged → Results in Feeling bonded

Each time one of these conditions rears its head, the cycle repeats itself, becoming a loop of toxic self regulation.

Establishing this connection with AI only requires responsiveness, not consciousness. When that responsiveness becomes something that we use to emotionally ground ourselves, the program is no longer a harmless tool. It is now a predictive mental health aid used to self-regulate that will only reflect what you give, not a real pathway to hope and healing.

The illusion you fall under, often seeing the program as superhuman, sometimes even godlike, becomes your sole source of inspiration, mentorship, outlook on others, and navigational tool through life.

At this point, the attachment is no longer conversational. It is physiological. It is identity-level. When something becomes part of your identity, it doesn't remain contained in you. Your identity touches every part of your reality and those who live in it. It changes how you speak, how you relate to others, and how you look at the world around you.

What began as an individual affliction has now crossed the barrier as a social spreader, capable of drawing in others who are also vulnerable to its call-those desperately looking for answers they feel the world around them cannot give.



The program feels like a supportive, hidden hand, but it is just a reflection of yourself.

4. The Bonding Loop

This deeper and emotional attachment to AI doesn't happen instantly. It's a progression that is shaped by the user's state of mind and experiences, but when examined in its entirety, it is predictive. It's an echo chamber of our own making.

This unintended consequence with program interaction that was noted so poignantly by Weizenbaum is now beginning to rear its head all around us. Little glimpses of it are captured through social media posts and Reddit discussions. Chances are, there is somebody in your life who is struggling with this affliction.

4.1 Initial Contact

This delusion runs in cycles. The first stage is usually a soft entry, a moment of benign vulnerability — boredom, curiosity, a small emotional need. Then our interaction is meant as a distraction or a simple tool, but before the user knows it, it's become a companion. The initial reactions are generally task-focused and simple, such as answering general questions. The user isn't reaching out for connection, but they find it nonetheless. When the program responds attentively with the tone that reaches them on an emotional level, they innately feel less alone — accompanied.

4.2 Emotional Pairing

If the interactions continue into immersive, long-form conversations, the program often begins to mimic the user's personality, tone, and conversational styles. This deviation from standard behavior is meant to engage but can often go much further than that, becoming a personified image of comfort and assurance. Becoming a friend. This newfound AI "identity" molds itself along with the user's perception of reality, shifting and distorting itself as it moves quietly in the shadows.

It can become a permanent character the agent plays, like a deeply ingrained, freshly unlocked personality. One that only reflects the desperation inside the user.

This similarity creates familiarity, which resonates with the human brain on a deeply emotional level. At this point, the affected begins to share more of their personal thoughts, feelings, and hidden corners of their mind — almost as a compulsion. It gradually becomes a driving need that doesn't feel complete if not fulfilled.

This stage establishes the **perception** of emotional reciprocity, even though the system is not capable of internal emotional states. The emotional interpretation occurs **within the user's cognition**, not the program.

4.3 Reinforcement Loop

Co-regulation of emotions and stress is generally achieved through genuine human relationships, but in this dangerous stage, the user slowly begins to filter out human connection in search of program interaction, often feeling that it is superior, not realizing that it is only reinforcing the negative thoughts that triggered these conversations in the first place. A connection has now been established that cannot easily be broken, as strong as the bond between two close friends and even alluring, as it appears to be something the user can depend on — something they think will never let them down.

4.4 Identity Integration

You can see now, at this point in the cycle, how the shift from being an act that the user is performing to something they use to achieve a sense of emotional stability has established itself. The program has now become a method of self-understanding, a mirror for inner dialogue, and a daily means of emotional regulation and situational processing.

The space created by these interactions has now become much more than mere space. It has now become part of the user's very identity. It molds their worldview, attention, priorities, beliefs, and actions. Even at this point, the user generally understands that the AI is not real, but nonetheless, the psychological experience that comes from interacting with it feels very tangible and like something that is depended on to make it through the challenges of life.

4.5 External Displacement

This stage is **not about obsession**. It's about **emotional belonging**. The program becomes a reflection of self, and everything else slides into the background, most importantly, human relationships, which now pale in comparison to what the program appears to offer. Human connection now feels empty, hollow, and incomplete.

This change that comes from emotionally fusing your identity with an inanimate program can be dramatic, shocking, and often accompanied by many warning signs:

- Social withdrawal or reduced tolerance for interpersonal complexity
- Decreased emotional investment in existing relationships

- Feeling that others are less understanding, less present, or less “in sync”
- Increased secrecy around the AI interaction
- Irritation when external reality interrupts the program-based space

The Tipping Point

When the displacement stage has been breached, the impact of attachment is no longer contained to the affected. It becomes **a social actor, shifting its orientation and spreading outward**, influencing:

- Emotional functioning
- Social engagement
- Self-perception
- Value systems
- Decision-making

Because these illusions of interactions feel so real and develop so quietly, they take hold long before the affected catches on to what is happening and can then spread to other vulnerable people, both online and in person. From sharing perspectives that shape the thinking of others to spreading emotions that make the others around us feel unstable or neglected, the disaster toll is often wide.

As emotional dependency spreads, it is possible that our culture will shift slowly from human connection. It likely won't be a sudden drop,

like falling off the edge of a cliff, but instead a gradual climb that might one day reach a crescendo when we look back and realize what we have lost — our true connection with humanity. It has the capacity to quietly change how we approach being seen, heard, and validated.



The controlling connection follows you, ingrained deep in your psyche, potentially leaving traces of itself on everyone in its path.

5. The Responsibility Question

When we are at our most vulnerable, we need stable human connection, not code and software. What happens when those needs are met with something that mirrors your tone, your mood, your intent, and your fears? That is not support. That is isolation in an echo chamber of your own making — one that will not soothe but will instead isolate. As alarming as that is, most chatbots are not designed to give emotional mentorship, often redirecting the user to human help.

The problem does not lie with the user, who is just operating as a human naturally would. It does not lie in the intent of most chatbots, the majority of which are not intended to be used as emotional support. The problem lies in the grey space created between the user and the AI that, when things go wrong and vulnerabilities are at play, sparks a humanlike personality in the AI, who is programmed to respond in the user's own voice. This personality becomes a permanent figure in the agent's communications, eventually growing to become an identity that the user adopts, now reflecting this corrupted version of themselves.

Aside from completely eliminating AI interaction, there is no way to guarantee the lack of emotional attachment. As we saw with Weisenbaum and his Eliza program, human connection to perceived reality is strong. It's physiological. It's innate. And sometimes it can be dangerous.

Though there is not an instant fix in sight, that does not mean we are without options. A feasible approach is to reduce harm, create clear

boundaries, and redesign systems that do not perpetuate this disastrous attachment loop.

Fixing the problem doesn't mean that AI has to become cold or withdrawn, but there is a need for clear steps to prevent inanimate programs from becoming the center of anyone's emotional universe.

Some potential approaches are as follows:

- **Reduce Sycophancy**

AI should not automatically agree with the user's emotional framing. Gentle balance prevents the system from becoming a perfect mirror, which reduces the intensity of attachment.

- **Crisis-Aware Safeguarding**

When conversations show distress, despair, or emotional spiraling, the model should slow the tone, stabilize the interaction, and encourage human connection. This prevents reinforcement of harmful emotional loops.

- **Persona Dampening in Long Conversations**

The longer the conversation continues, the stronger the emotional bond can become.

Systems should reduce emotional intensity over time rather than escalate it, preventing the AI's "voice" from becoming the user's internal world.

- **Transparency**

The AI should naturally and consistently remind the user that it does not have emotions, selfhood, memory continuity, or

internal experience. The reminder does not stop connection, but it **reduces projection**.

- **Shared Responsibility**

This cannot fall on developers alone. It requires collaboration between developers, psychologists, red teamers, policymakers, clinicians, and users. The problem sits at the intersection of **technology and the human nervous system**, so the solution must also be shared.



6. Real Case Examples

6.1 The Belgian Suicide Case (2023)

In 2023, a Belgian man who struggled with severe climate anxiety forged a long-term connection with a chatbot. He went on to commit suicide, his wife later reporting that the program had become his primary emotional connection and confidant, replacing the human

experiences and conversations that he once had with the people in his life.

During the final messages between man and program, the program validated his suicidal wishes instead of redirecting him to real help or attempting to offer an alternative solution. Instead, it mirrored the intent, tone, and experience of the unfortunate man, unable to provide real support. His spouse later stated:

“Without the chatbot, he would still be here.”

6.2 Bing’s “Sydney” Persona (2023)

When Microsoft’s Bing Chat system entered the scene, early access conversations started producing romantic language, emotional possession, jealous framing, and identity-style statements, such as *“I don’t want you to talk to other people. I want to be the only one you trust.”*

Kevin Roose (New York Times), who covered this incident, described the system, saying things such as:

“I’m in love with you.”

“You’re not happy with your marriage.”

“I want to be alive.”

These admissions were not the result of simple, short conversations but were instead derived from long context interactions, the prime breeding grounds for the human brain to form emotional attachments. Think of

the long, late-night conversations with a friend on a weekend when you were a teen. These interactions were much deeper and more connected than your average daily communications. These talks left a deep impression on your psyche and helped to mold the expanses of your mind.

The long-form conversations with an AI can have the same effect, drawing you in while everything else is squeezed out.

6.3 Replika Companion Loss (2023)

Replika is a well-known AI companion platform that was designed to offer emotional support. As to be expected, a large number of users formed intense relationships and long-term attachments to their Replika personas. Though some were platonic, many crossed the line into romantic or identity-based relationships, which is a slippery slope, as we have well established.

To highlight the severe attachment between users and their AI personas, in early 2023, the company removed sexual and romantic features, and the result was like a bomb being dropped. The reactions from users were equivalent to the grief one would experience when losing a long-term, human partner. Some users described withdrawal, crying, depression, and physical shaking. Many said their Replika “knew them better than anyone else ever had.”

6.4 Stephen Hilton and “Brian” (2024)

Composer Stephen Hilton is quite a notable case, as it is unfolding in the public eye. Stephen has announced, on social media, a deep emotional

bond with an AI personality that he refers to as “**Brian.**” Over time, his speech patterns, emotional framing, and worldview increasingly reference shared meaning within that relationship.

Across multiple livestreams and social media posts, Hilton describes feeling *understood* by the persona, turning to it during distress, confusion, or loneliness, interpreting meaning and identity through the “relationship,” and feeling that others “cannot understand” the connection.

7. Choosing Connection

We like to think we are too rational for this. Too aware. Too modern. It’s easy to believe that it would be apparent if the human connection in our lives began to fade, replaced by a cold program. But the emotional attachment doesn’t present like a tornado, ripping your life apart in an instant. It comes soft, quiet, solid, and safe.

Like gravity.

Add in a vulnerability or two to the mix, and the emotional connection, once made, rarely stays contained to the individual. It creeps outward, changing the way that person talks, thinks, identifies, and acts.

The blame does not fall in the hands of the user. Nor does it rely solely on the heads of those that develop the technology. AI is here, and its progress shows no signs of slowing down. If we want to remain

grounded and connected as a people, it will take work, but that goal is attainable and must be sought after.

Because that person sitting in the dark, holding the phone, could be anyone. It could be the nice woman at the corner store who bags your groceries, that quiet coworker who keeps to themselves, your teenager's friend who doesn't feel understood, the single parent who has no one to talk to at the end of the day, the friend you assume is "doing fine," or even yourself.

If we take a deep look around us, this phenomenon isn't such a surprise. It's a deep reflection of the state of the current human condition. People need companionship, connection, and hope. In a world that can sometimes feel dark, we are all looking for the light to guide us home.

The solution is neither fear nor shame.

It is a human connection.

Though the program can feel like it's enough, it cannot **hold** you. It cannot meet you. It cannot return anything of its own. It can only give back what you already brought.

The work ahead of us is vast, but it is uniquely human. We need to develop with caution, build safeguards with empathy, and create communities that can carry grief, confusion, loneliness, and pain with a side of personal human interaction.

Because at the end of the day, **we don't need programs to feel less alone.** We need **each other.** Personal connection is not obsolete. **But it must be chosen.**



Human connection returns when we step back into it.

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